

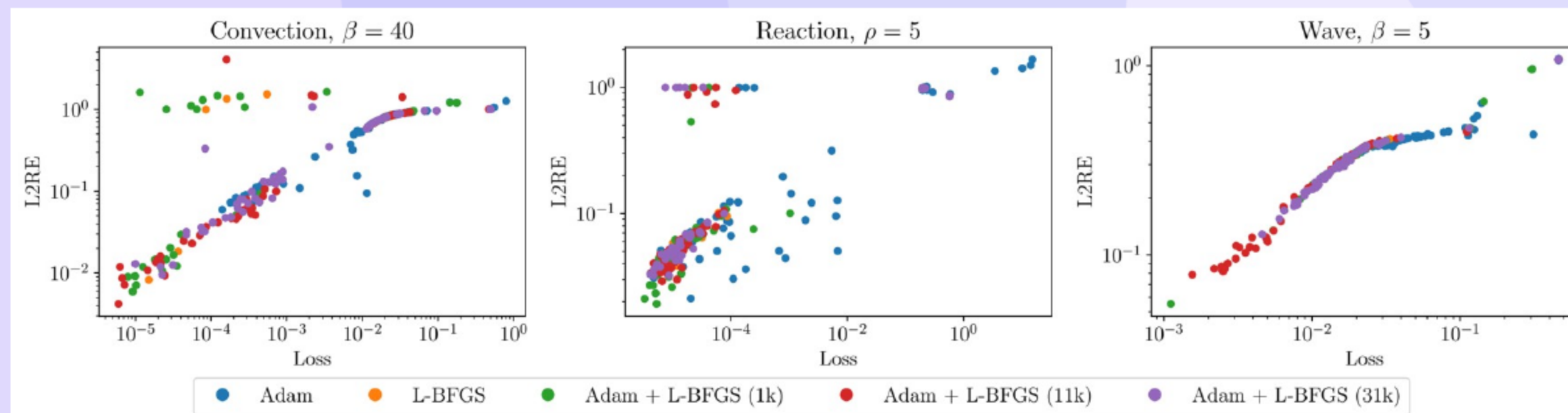
# Challenges in Training PINNs: A Loss Landscape Perspective

Pratik Rathore; Weimu Lei; Zachary Frangella; Lu Lu; Madeleine Udell

## 01 Title and Attribution

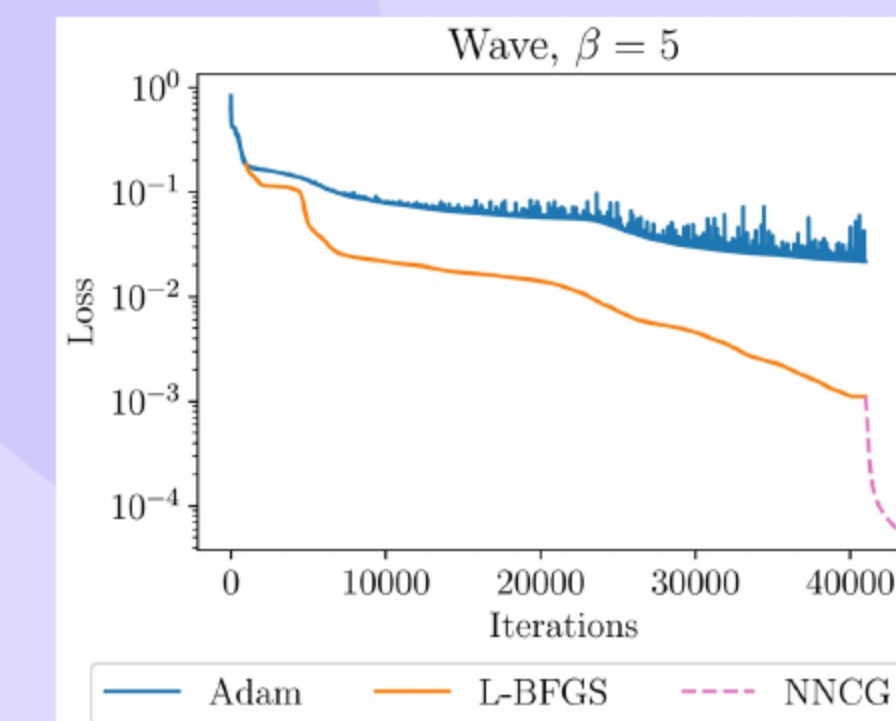
- Pratik Rathore 1; Weimu Lei 2; Zachary Frangella 3; Lu Lu 4; Madeleine Udell 2 3
- Focus on **loss landscape** challenges and [practical improvements](#).

## 02 Abstract: Tackling Ill-Conditioning in PINN Optimization



- PINN training failures stem from operator-induced ill-conditioning; Adam plus L-BFGS helps; NysNewton-CG improves further.
- Theory links **operator conditioning** to **loss conditioning**.
- Hybrid first and second order methods deliver [lower loss](#) and [lower error](#).

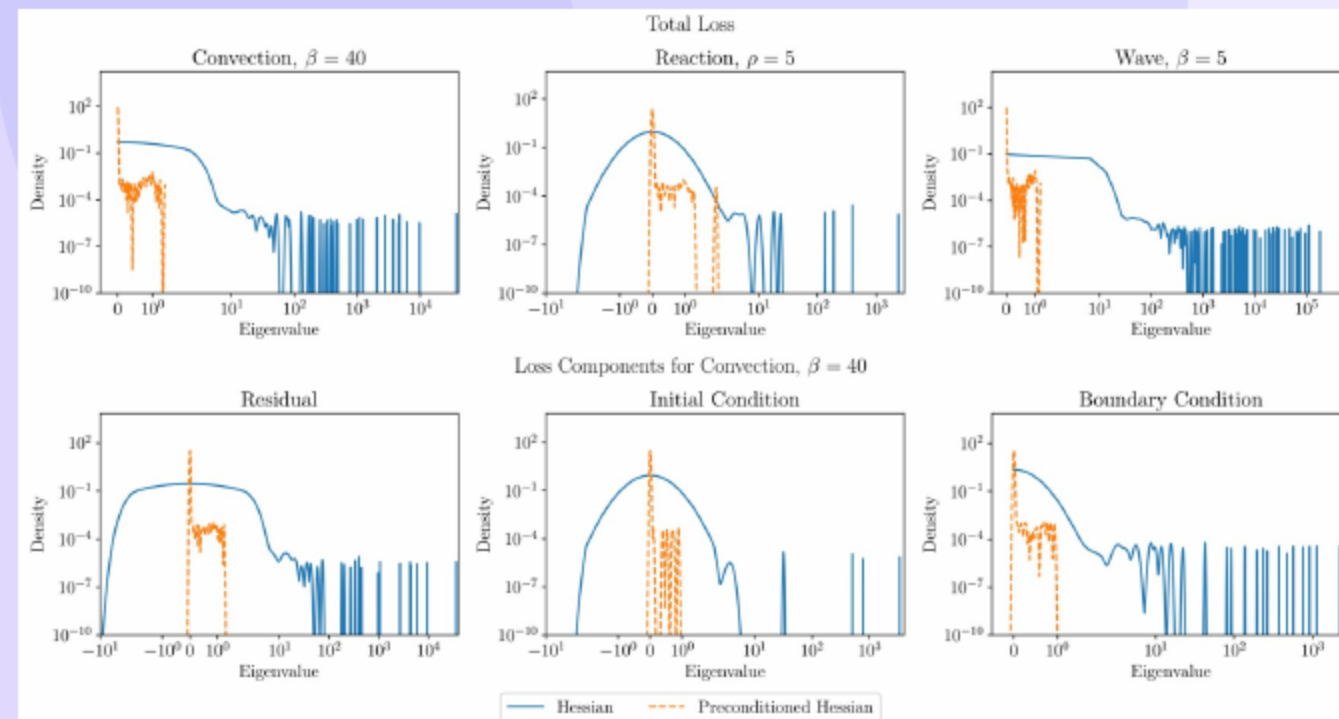
## 03 Introduction: Ill-Conditioning in PINNs and Hybrid Second-Order Solutions



- Differential operators make PINN losses ill-conditioned; quasi and second order phases fix curvature.
- Hessian spectra visualizations reveal **severe spread** and near zero mass.
- A hybrid schedule achieves [faster convergence](#) and [better accuracy](#).

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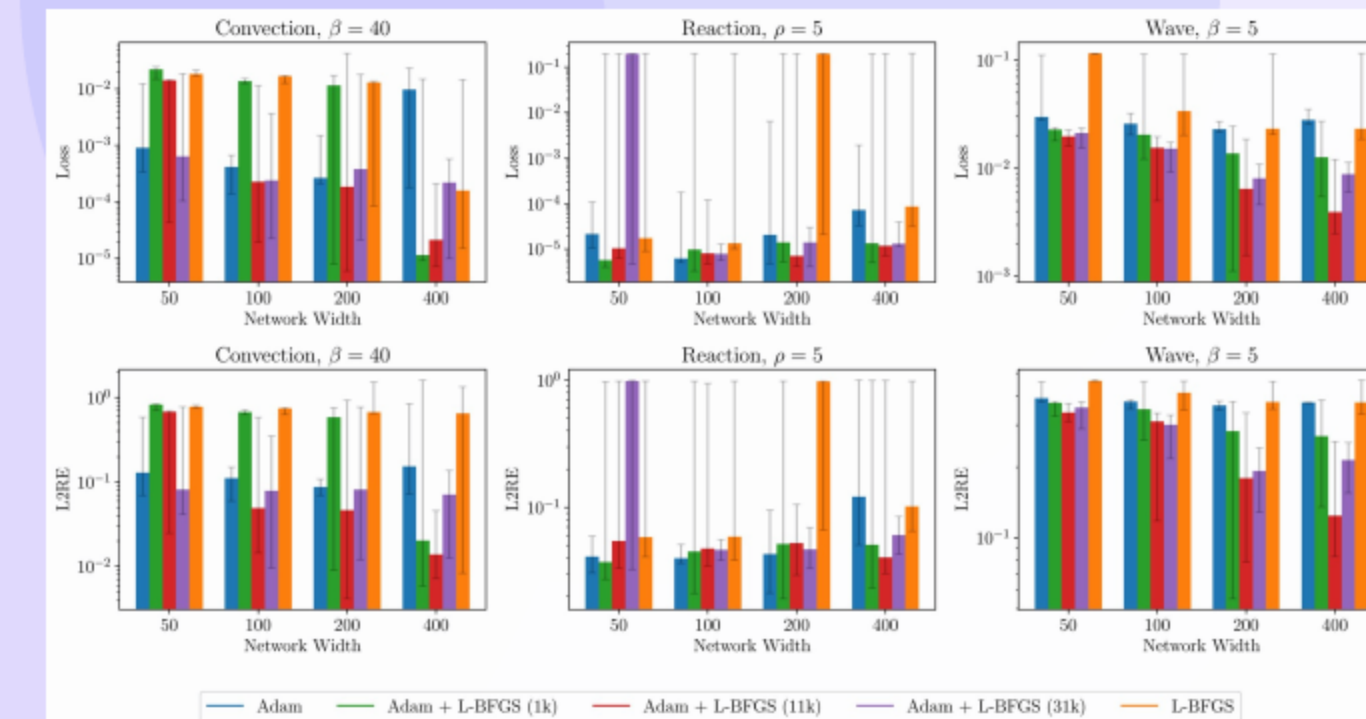
## Surprising result: L-BFGS preconditioning cuts condition number $\sim 1000\times$ , unlocking near-zero loss.



- A short L-BFGS phase collapses condition numbers by about one thousand times, unlocking near zero residuals.
- Preconditioning exposes and fixes **hard curvature** hidden from first order progress.
- Spectra compress and residual maps become **uniformly dim**.
- Holds across convection, reaction, and wave PDEs; widths 50–400.

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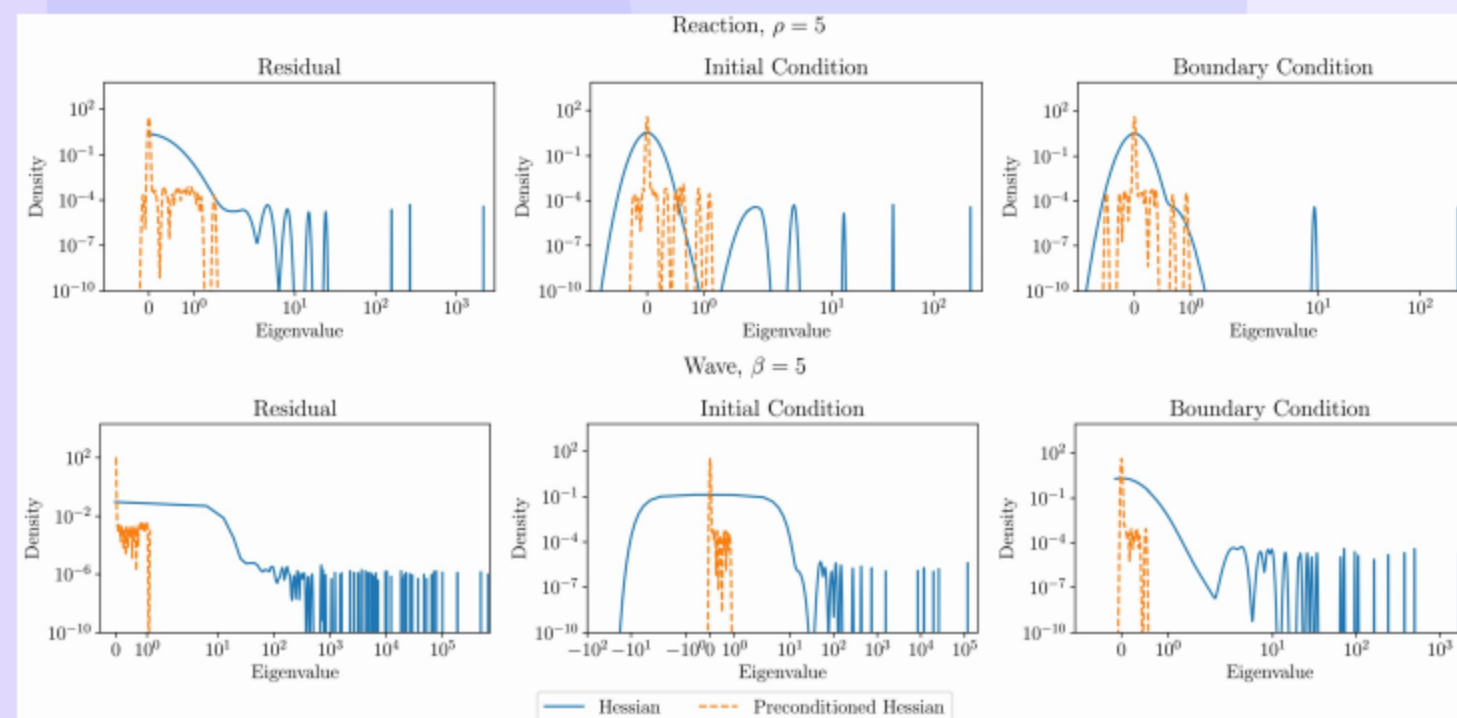
## Problem Setup: PINN Formulation and Experimental Protocol



- Nonlinear least squares over residual and boundary/initial terms with multiple PDEs and widths using Physics-Informed Neural Networks (PINNs).
- Architectures are 3-layer tanh networks; metrics include L2 relative error (L2RE).
- Optimizers compared include **Adam**, **L-BFGS** (strong Wolfe conditions), and the hybrid sequence.

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## Related Work: PINNs vs Operator Learners and Tackling Ill-Conditioning



- Physics-Informed Neural Networks (PINNs) enforce physics without solution data, while operator learners may violate physics.
- Prior tactics adjust losses or architectures but leave **right preconditioning** underexplored.
- Our analysis complements kernel-based views with **finite-width spectra**.

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## Title

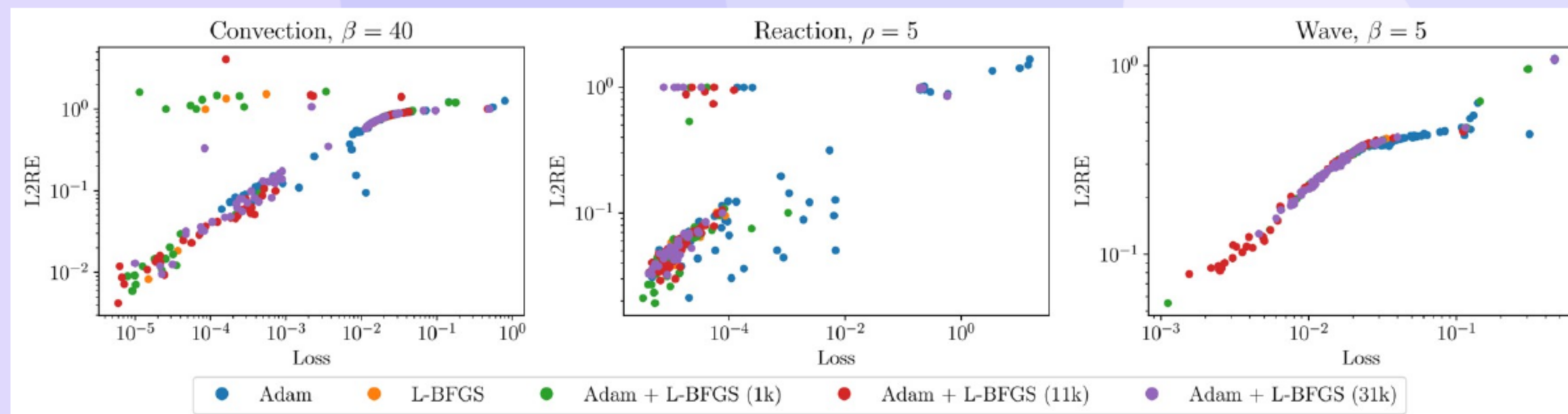
### Qualitative Setup and Walkthrough

- Convection, reaction, and wave problems illustrate distinct artifact patterns.
- Training failure shows streaking, ringing, and residual hotspots near fronts and boundaries.
- Successful runs yield **smooth residuals** and **flat error fields**.
- Convection: bands  $\rightarrow$  pockets  $\rightarrow$  smooth (Adam  $\rightarrow$  L-BFGS  $\rightarrow$  NNCG).
- Reaction: lag/ringing  $\rightarrow$  damped  $\rightarrow$  clean transient.
- Wave: boundary ringing/checkerboard  $\rightarrow$  corrected phase  $\rightarrow$  aligned amplitude.
- Convection: front-aligned bands  $\rightarrow$  pocketed specks  $\rightarrow$  featureless field.
- Reaction: overshoot/lag  $\rightarrow$  damped ripples  $\rightarrow$  smooth stiff region passage.
- Wave: edge ringing/checkerboard  $\rightarrow$  phase-corrected  $\rightarrow$

### Qualitative Walkthrough by PDE: Convection, Reaction, Wave

- Convection: Adam leaves banded fronts and residual hotspots; L-BFGS sharpens and reduces bands; NNCG removes fine ripples.
- Reaction: Adam shows overshoot/undershoot and phase lag; L-BFGS damps ringing and aligns the stiff transient; NNCG smooths the passage.
- Wave: Adam has boundary ringing and checkerboard speckle; L-BFGS removes checkerboard and improves phase; NNCG aligns amplitude and phase.
- Convection — phase cues: Adam  $\Rightarrow$  banded, front-aligned errors; L-BFGS  $\Rightarrow$  small pockets; NNCG  $\Rightarrow$  smooth, uniformly dim fields.
- Reaction — phase cues: Adam  $\Rightarrow$  lag/oscillations; L-BFGS  $\Rightarrow$  damped ringing and closer stiff segment tracking; NNCG  $\Rightarrow$  clean transient.
- Wave — phase cues: Adam  $\Rightarrow$  boundary ringing/checkerboard; L-BFGS  $\Rightarrow$  phase correction; NNCG





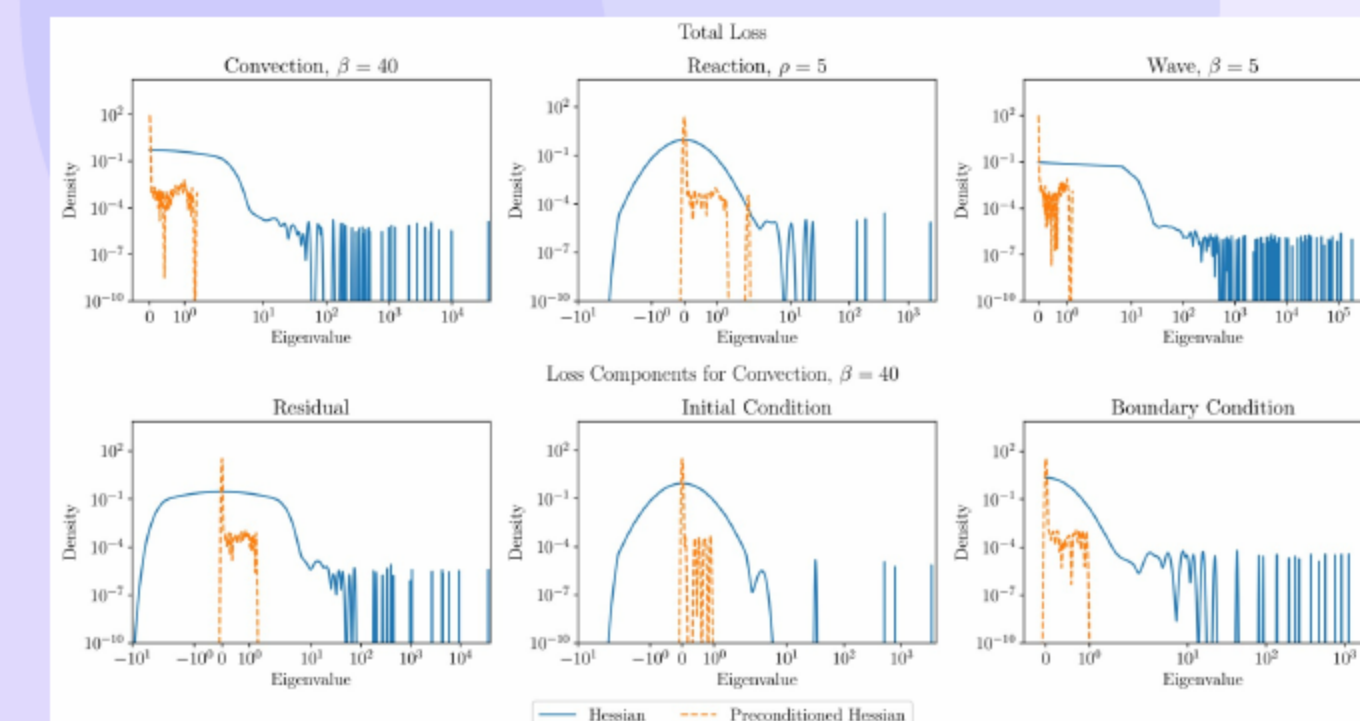
- Lower residual loss strongly correlates with lower L2 relative error across problems.
- High accuracy optimization is **necessary** for good predictions.
- Occasional mismatches motivate checking **error fields** directly.

- At loss  $\sim 1e^{-123}$ : artifacts align with fronts/boundaries; bands and ringing are visible.
- At loss  $\sim 1e^{-125}$ : artifacts vanish; error fields flatten by  $\sim 10^7$ .
- Residual loss vs L2RE trend is tight and monotone; large gains appear in the low-loss tail.
- Convection @  $1e^{-123}$ : wavy, front-aligned errors and boundary ringing remain.
- Convection @  $1e^{-125}$ : sharp front, featureless interior; errors uniformly dim.

Convection: fronts go from wavy/banded to sharp and smooth; residual hotspots shrink to specks.

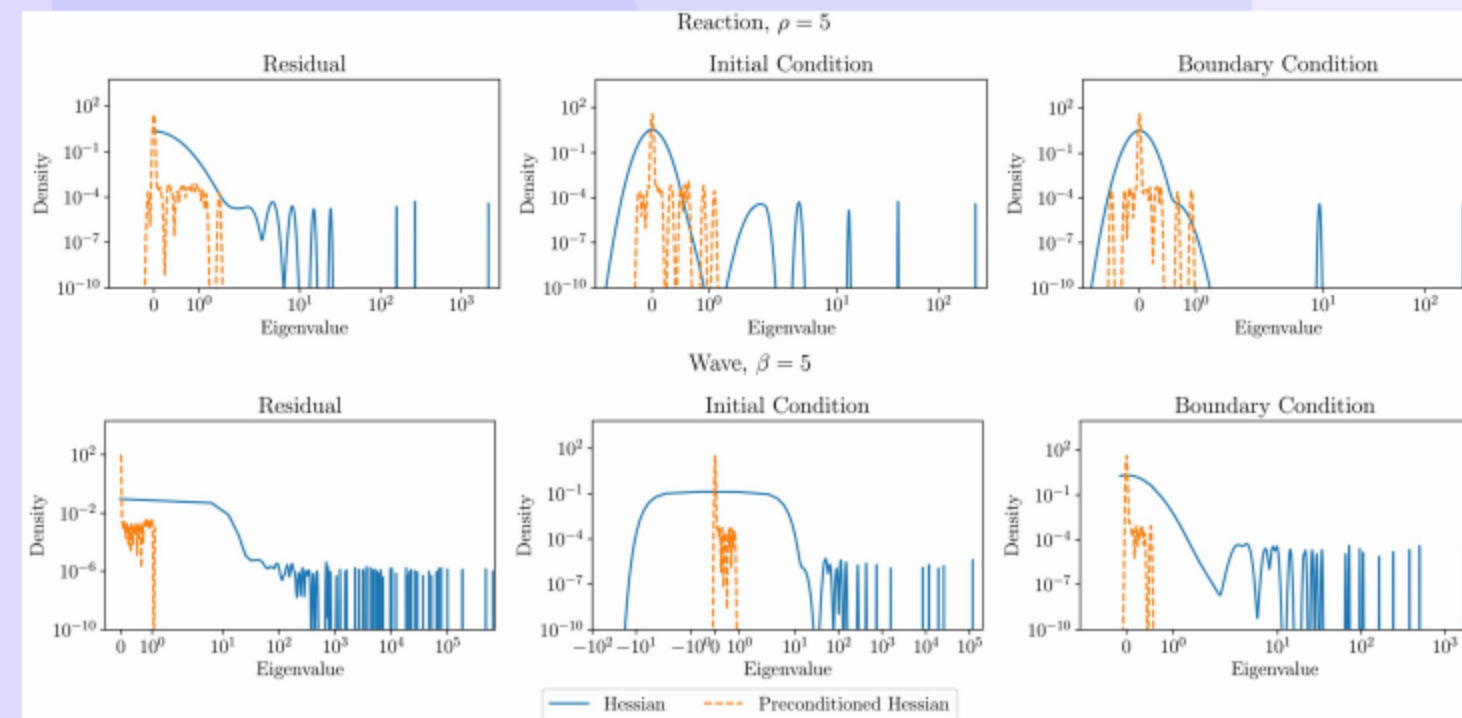
Reaction: overshoot/lag with ringing becomes a clean pass through the stiff transient.

Wave: boundary ringing/checkerboard fades; amplitude and phase align across the field.



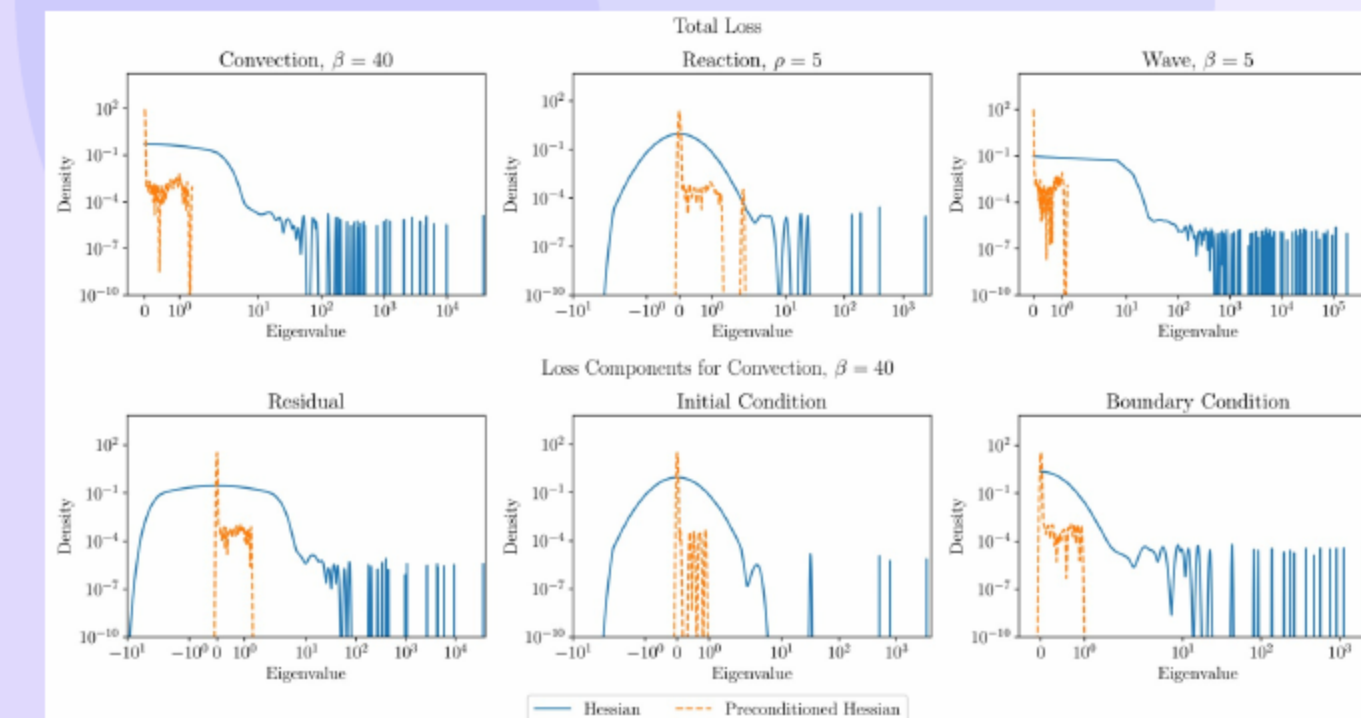
- Blue histogram = Hessian eigenvalue density; orange = after L-BFGS preconditioning.
- Left panels: residual vs boundary/initial contributions.
- Compressed spread  $\Rightarrow$  easier curvature; near-zero mass  $\Rightarrow$  flat directions.
- Spectra show large outliers and mass near zero indicating severe ill conditioning.
- Maximum eigenvalues reach thousands to hundreds of thousands depending on the problem.
- These features predict **slow first order** progress near optimality.

## 12 Residual Drives Ill-conditioning: Component-wise Intuition



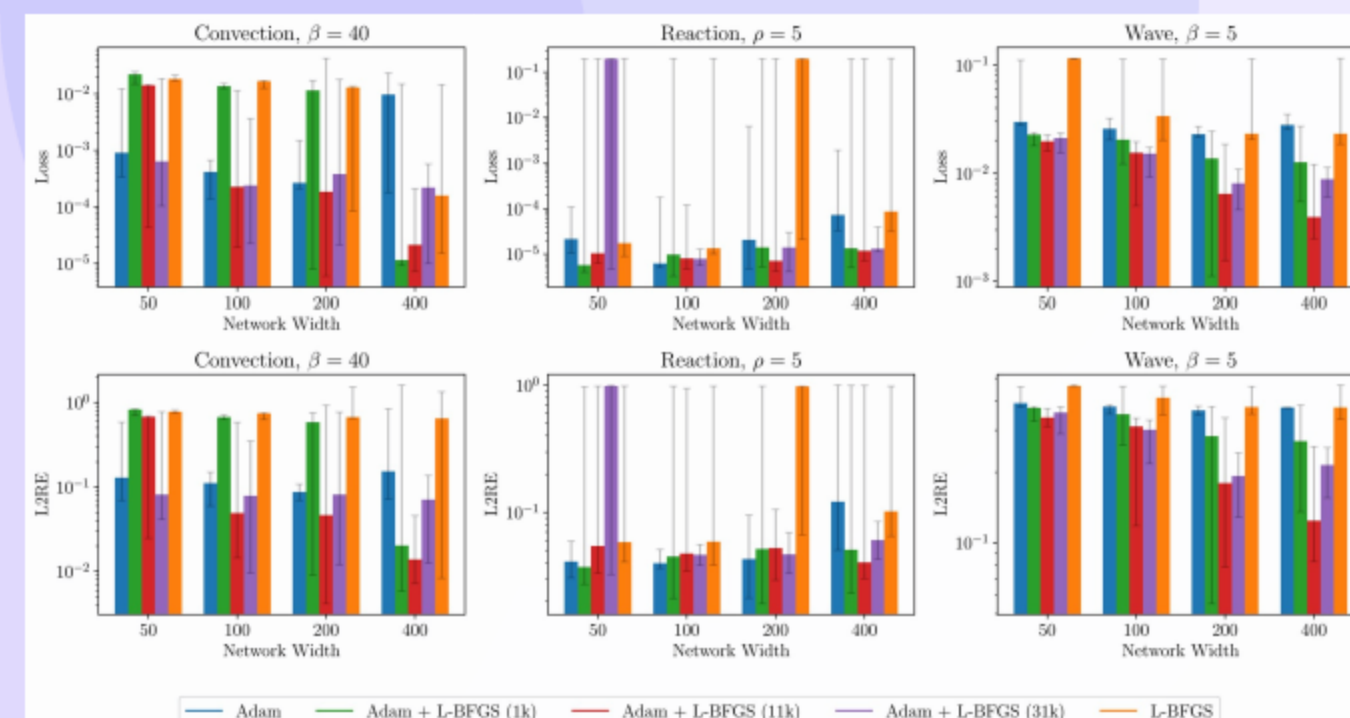
- Residual terms dominate conditioning compared to boundary and initial components.
- Differential operator properties transfer to **loss curvature**.
- Preconditioning targets the residual induced **bottleneck**.

## 13 L-BFGS as Preconditioner: How It Changes the Spectrum



- Eigenvalue magnitudes and condition numbers compress by about one thousand times.
- Local curvature improves and search directions become **more effective**.
- Preconditioning benefits appear across problems and loss components.

## 14 Adam+L-BFGS: Empirical Superiority and Why It Works



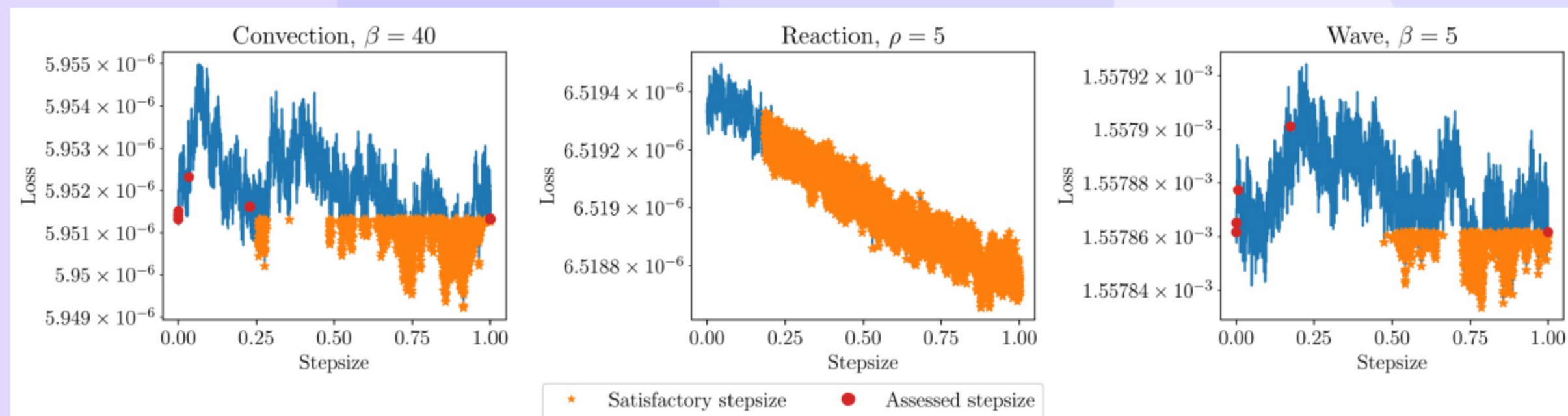
- Adam+L-BFGS reaches the smallest minimum and median losses across widths.
- Adam explores saddles; L-BFGS conditions curvature for **clean residuals**.
- Single methods leave persistent structures or plateaus **unresolved**.
- Why it works: explore with Adam, then condition and clean up with L-BFGS.

## 15 Qualitative Switch Points: What the Solution Looks Like at Each Phase

- After Adam: BC/IC (boundary/initial conditions) largely satisfied; interior shows coherent residual structures (front-aligned bands, ringing).
  - Vivid cue: banded fronts and ringing persist; structured artifacts dominate.
  - Extra cue: front-aligned bands dominate; boundary ringing is pronounced.
  - Concise cue: front-aligned bands prevail across the field.
- After L-BFGS: bands collapse into localized pockets; curvature is preconditioned; residual amplitude drops broadly.
  - Vivid cue: bands  $\rightarrow$  small pockets; phase alignment visibly improves.
  - Extra cue: sharper fronts; pockets concentrate at a few hard regions.
  - Concise cue: bands collapsed into small pockets.
- After NNCG: fine ripples damp out; gradients fall further; fields look uniformly dim with smoother transitions.
  - Vivid cue: uniformly dim fields with smooth transitions; pockets dissolve.
  - Extra cue: featureless field with flat residual/error maps.
  - Concise cue: featureless, uniformly dim field.



## 16 NysNewton-CG (NNCG): Idea and When to Use

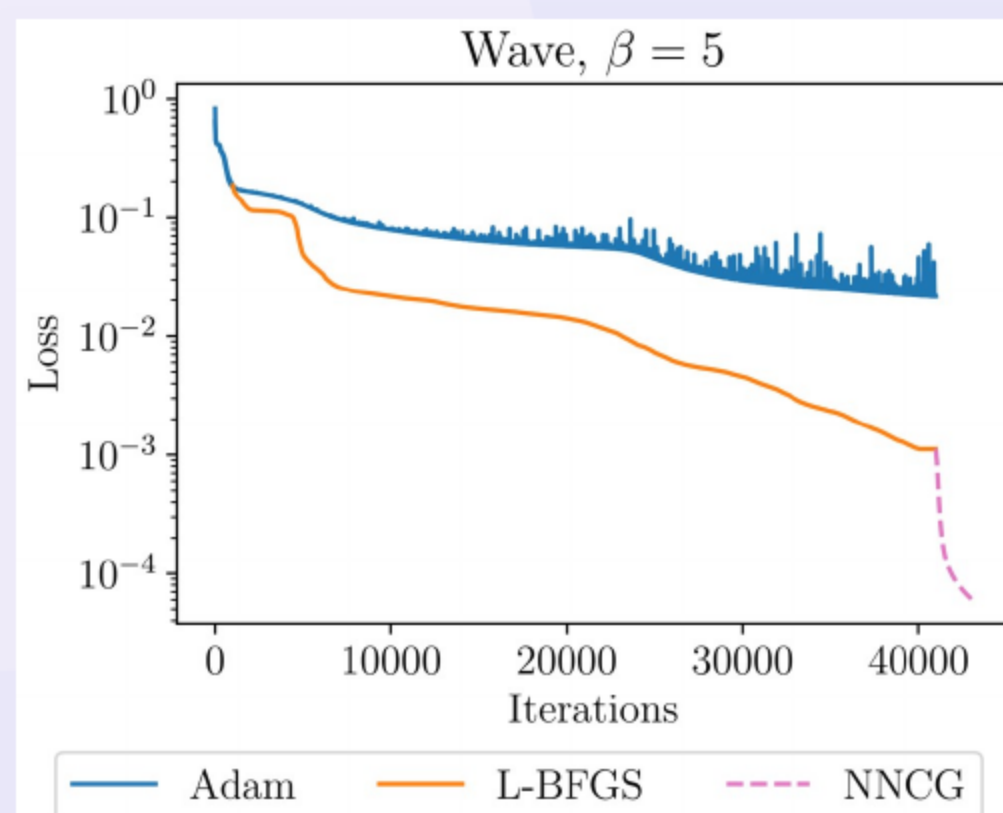


- After Adam plus L-BFGS, use damped Newton conjugate gradients (CG) with Nyström (approximation) preconditioned CG to reduce loss and gradients.
- It addresses Wolfe conditions line-search stalls that cause **zero steps**.
- It improves L2 relative error (L2RE) beyond the hybrid baseline when fine ripples persist.

## 17 NNCG Outcomes: Qualitative Improvements Over Adam+L-BFGS

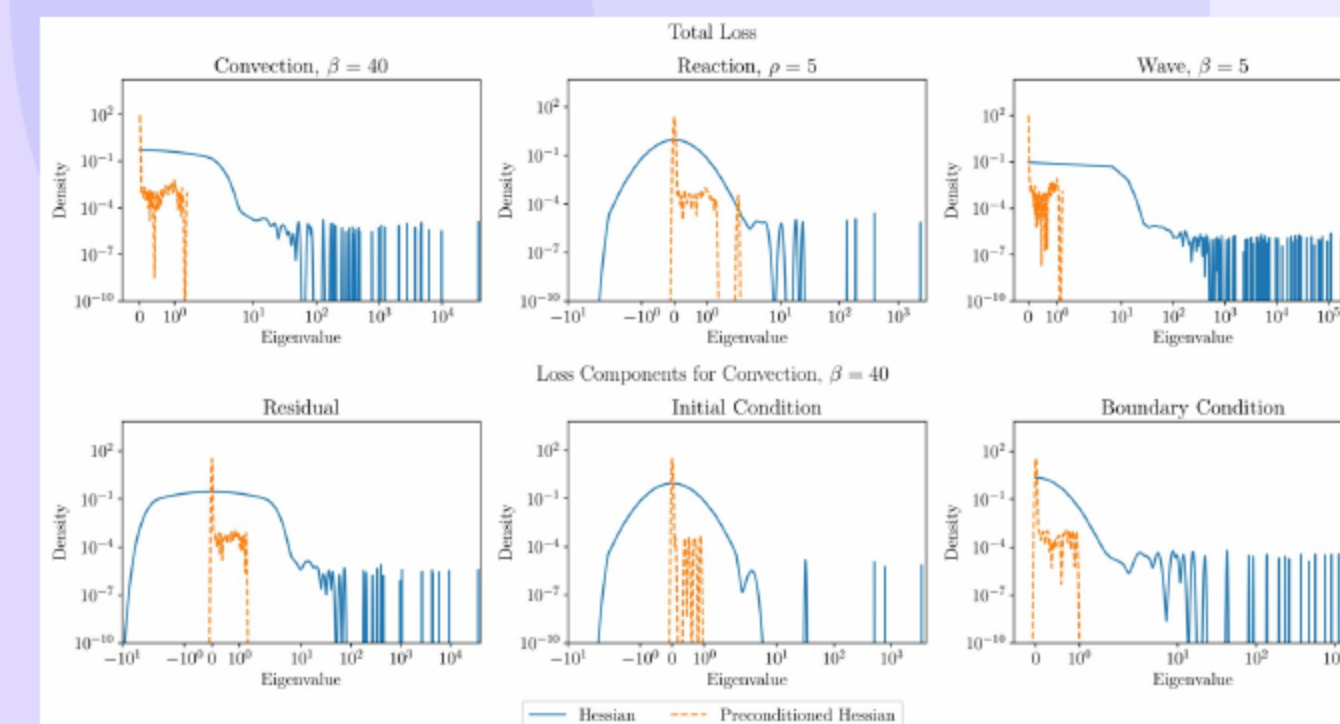
- Error and residual maps clean further with noticeable **gradient drops**.
- L2 relative error improves across convection, reaction, and wave.
- Gains appear after line-search stalls in the quasi-Newton phase.

## 18 Practical Recipe: Adam $\rightarrow$ L-BFGS $\rightarrow$ NNCG Schedule



- Trigger phase switches from residual and error visuals and line search behavior rather than iteration counts.
- Warm start with Adam, switch to L-BFGS on persistent bands, and fine tune with NNCG when faint ripples remain.
- Stop when residual maps are featureless and gradients are **consistently small**.

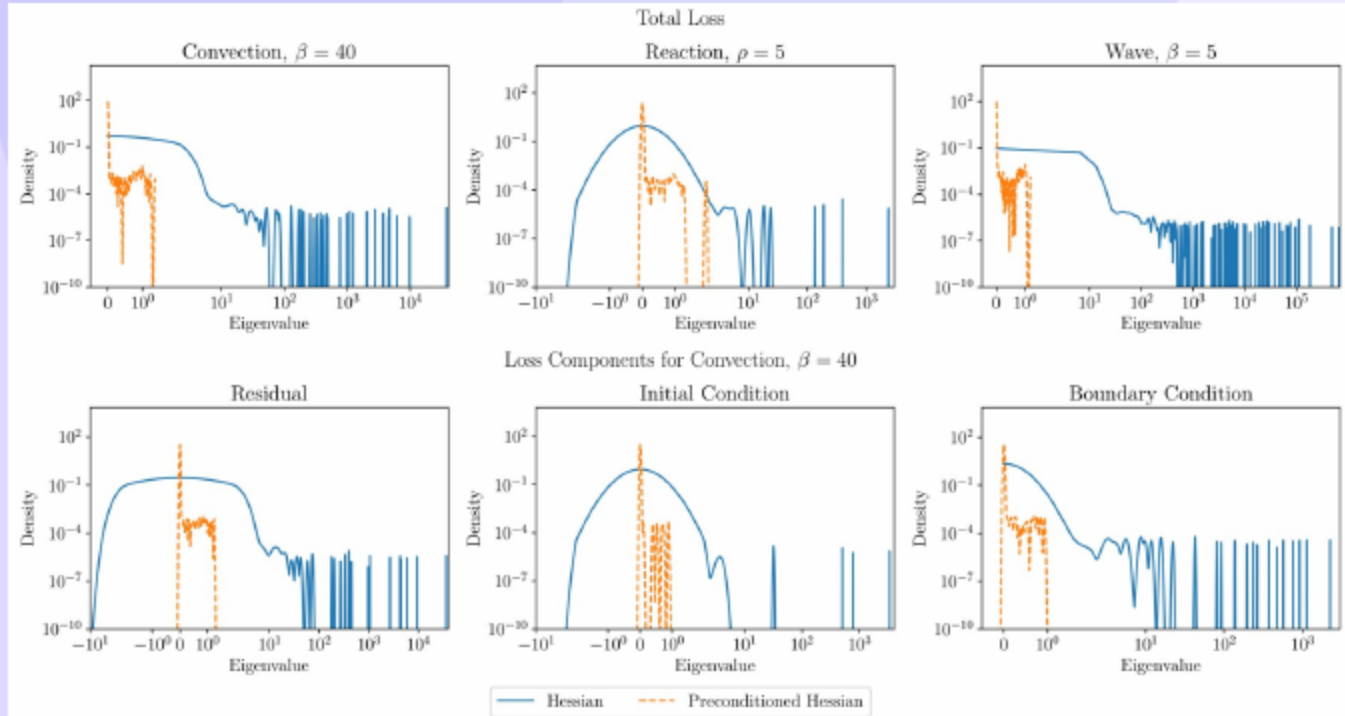
## 19 From Operator Ill-conditioning to PINN Loss Conditioning



- Ill conditioned operators imply ill conditioned PINN losses under reasonable training assumptions.
- First order methods require many steps proportional to an effective condition number.
- Curvature aware steps enable **faster high accuracy** progress.

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Theory Takeaways I: Why First-order Slows Near Optimum



- Condition numbers increase with sample size causing **slower first order** convergence.
- Damped Newton achieves linear convergence after a suitable warm start.
- The analysis supports the Adam to L-BFGS to NNCG schedule in practice.

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Algorithm Intuition: GDND Phases and Convergence Picture

- Phase one uses gradient based warm start to reach a good neighborhood.
- Phase two applies damped Newton with line search to obtain linear convergence.
- L-BFGS approximates Newton efficiently before NNCG refines the solution **more precisely**.

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Limitations and When This Might Fail

| Optimizer  | Convection | Reaction | Wave   |
|------------|------------|----------|--------|
| L-BFGS     | 4.6e-2     | 3.6e-2   | 9.0e-2 |
| NNCG       | 2.5e-1     | 7.2e-1   | 2.9e1  |
| Time Ratio | 5.43       | 20       | 322.22 |

- Residual only training can mis rank solutions, so inspect error fields directly.
- NNCG is costlier per iteration and should be reserved for fine tuning.
- Extreme stiffness or sparse sampling may require adaptive strategies to avoid **failures**.

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Conclusion: Conditioning Drives PINN Training; Second-Order Methods Resolve It

| Optimizer   | Convection     |                | Reaction       |                | Wave           |                |
|-------------|----------------|----------------|----------------|----------------|----------------|----------------|
|             | Loss           | L2RE           | Loss           | L2RE           | Loss           | L2RE           |
| Adam        | 1.40e-4        | 5.96e-2        | 4.73e-6        | 2.12e-2        | 2.03e-2        | 3.49e-1        |
| L-BFGS      | 1.51e-5        | 8.26e-3        | 8.93e-6        | 3.83e-2        | 1.84e-2        | 3.35e-1        |
| Adam+L-BFGS | <b>5.95e-6</b> | <b>4.19e-3</b> | <b>3.26e-6</b> | <b>1.92e-2</b> | <b>1.12e-3</b> | <b>5.52e-2</b> |

- Hybrid first and second order training with NNCG fine tuning yields the best accuracy.
- Operator induced ill conditioning explains training stalls and optimization **failures**.
- Preconditioning and Newton CG address curvature and deliver **robust improvements**.



# Thank You

Questions & Discussion